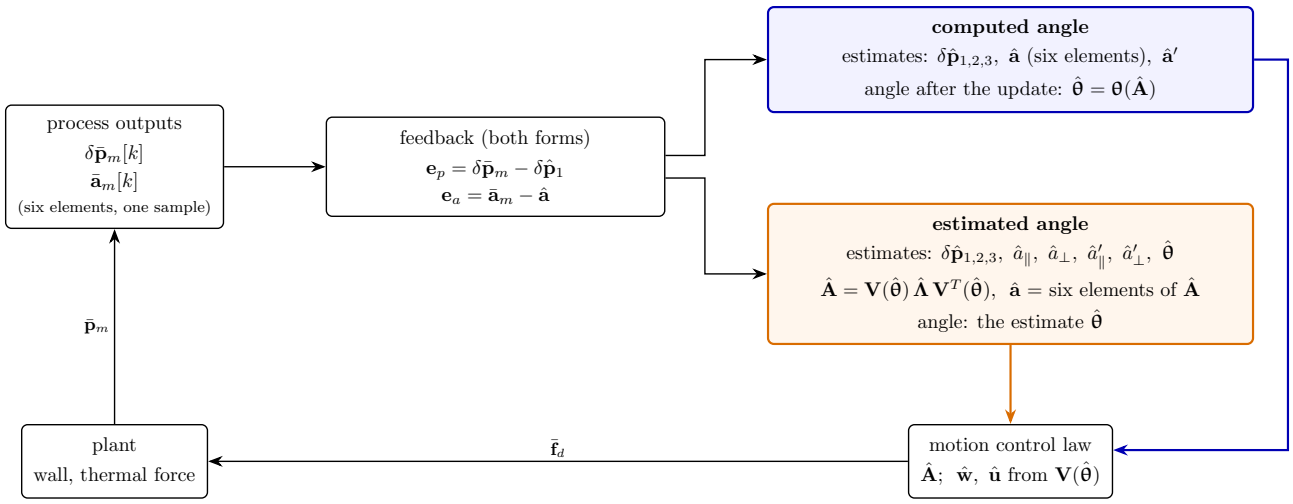
Computed angle and estimated angle of a tilted wall

2026-10-25 measurement coordinate (x, y, z), local coordinate (u, v, w)



1 Wall and motion gain matrix

bar: normalized (length $/R$, gain $/a_{\text{nom}}$); $\parallel$, $\perp$: parallel, perpendicular to the wall

$$\mathbf{V}(\boldsymbol{\theta}) = \begin{bmatrix} \cos\theta_y & \sin\theta_y \sin\theta_x & \sin\theta_y \cos\theta_x \\ 0 & \cos\theta_x & -\sin\theta_x \\ -\sin\theta_y & \cos\theta_y \sin\theta_x & \cos\theta_y \cos\theta_x \end{bmatrix} = [\,\mathbf{u} \;\; \mathbf{v} \;\; \mathbf{w}\,], \qquad \boldsymbol{\theta} = [\,\theta_x \;\; \theta_y\,]^T$$

$$\mathbf{w} = [\,\sin\theta_y \cos\theta_x \quad -\sin\theta_x \quad \cos\theta_y \cos\theta_x\,]^T$$

$$\bar{\boldsymbol{\Lambda}} = \operatorname{diag}\big(\bar{a}_\parallel,\ \bar{a}_\parallel,\ \bar{a}_\perp\big)$$

$$\bar{\mathbf{A}} = \mathbf{V}\,\bar{\boldsymbol{\Lambda}}\,\mathbf{V}^T = \bar{a}_\parallel\,\mathbf{I} + \big(\bar{a}_\perp - \bar{a}_\parallel\big)\,\mathbf{w}\,\mathbf{w}^T$$

elements of $\bar{\mathbf{A}}$ and their vector

$$\bar{\mathbf{A}} = \begin{bmatrix} \bar{a}_{xx} & \bar{a}_{xy} & \bar{a}_{xz} \\ \bar{a}_{xy} & \bar{a}_{yy} & \bar{a}_{yz} \\ \bar{a}_{xz} & \bar{a}_{yz} & \bar{a}_{zz} \end{bmatrix}$$

$$\bar{\mathbf{a}} = [\,\bar{a}_{xx} \;\; \bar{a}_{yy} \;\; \bar{a}_{zz} \;\; \bar{a}_{xy} \;\; \bar{a}_{xz} \;\; \bar{a}_{yz}\,]^T$$

change along the surface normal, h = distance to the wall

$$\bar{\mathbf{A}}' = \frac{\mathrm{d}\bar{\mathbf{A}}}{\mathrm{d}h}, \qquad \bar{\mathbf{a}}' = \frac{\mathrm{d}\bar{\mathbf{a}}}{\mathrm{d}h}$$

2 Process

hat: estimate, normalized as the bar quantity

$$\bar{\mathbf{p}}[k+1] = \bar{\mathbf{p}}[k] + \bar{\mathbf{A}}\left(\bar{\mathbf{f}}_d[k] + \bar{\mathbf{f}}_T[k]\right)$$
$$\delta\bar{\mathbf{p}}[k] = \bar{\mathbf{p}}_d[k] - \bar{\mathbf{p}}[k]$$
$$\bar{\mathbf{p}}_m[k] = \bar{\mathbf{p}}[k-d] + \bar{\mathbf{n}}_p[k]$$
$$\delta\bar{\mathbf{p}}_m[k] = \bar{\mathbf{p}}_d[k-d] - \bar{\mathbf{p}}_m[k]$$
$$\hat{\mathbf{p}}[k] = \bar{\mathbf{p}}_m[k] + \sum_{i=1}^{d} \hat{\mathbf{A}}[k-i]\,\bar{\mathbf{f}}_d[k-i]$$
$$\bar{\mathbf{f}}_d[k] = \hat{\mathbf{A}}^{-1}[k]\left\{\bar{\mathbf{p}}_d[k+1] - \hat{\mathbf{p}}[k] - \lambda_c\left(\bar{\mathbf{p}}_d[k] - \hat{\mathbf{p}}[k]\right)\right\}$$

dynamics of tracking error

$$\delta\bar{\mathbf{p}}[k+1] = \lambda_c\,\delta\bar{\mathbf{p}}[k] - \left(\bar{\mathbf{A}} - \hat{\mathbf{A}}[k]\right)\bar{\mathbf{F}}_{dp}[k] - \boldsymbol{\varepsilon}_p[k]$$
$$\bar{\mathbf{F}}_{dp}[k] = \bar{\mathbf{f}}_d[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{\mathbf{f}}_d[k-i]$$
$$\boldsymbol{\varepsilon}_p[k] = \bar{\mathbf{A}}\,\bar{\mathbf{f}}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{\mathbf{A}}\,\bar{\mathbf{f}}_T[k-i] - (1-\lambda_c)\,\bar{\mathbf{n}}_p[k]$$

F_x, F_y, F_z: elements of $\bar{\mathbf{F}}_{dp}[k]$

$$\left(\bar{\mathbf{A}} - \hat{\mathbf{A}}\right)\bar{\mathbf{F}}_{dp} = \begin{bmatrix} (\bar{a}_{xx} - \hat{a}_{xx})\,F_x + (\bar{a}_{xy} - \hat{a}_{xy})\,F_y + (\bar{a}_{xz} - \hat{a}_{xz})\,F_z \\ (\bar{a}_{xy} - \hat{a}_{xy})\,F_x + (\bar{a}_{yy} - \hat{a}_{yy})\,F_y + (\bar{a}_{yz} - \hat{a}_{yz})\,F_z \\ (\bar{a}_{xz} - \hat{a}_{xz})\,F_x + (\bar{a}_{yz} - \hat{a}_{yz})\,F_y + (\bar{a}_{zz} - \hat{a}_{zz})\,F_z \end{bmatrix} = \mathbf{G}_a[k]\left(\bar{\mathbf{a}} - \hat{\mathbf{a}}\right)$$
$$\mathbf{G}_a[k] = \begin{bmatrix} F_x & 0 & 0 & F_y & F_z & 0 \\ 0 & F_y & 0 & F_x & 0 & F_z \\ 0 & 0 & F_z & 0 & F_x & F_y \end{bmatrix}$$

change of the motion gain matrix with the motion along the surface normal

$$\mathbf{M}[k] = \Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,\delta\bar{\mathbf{p}}_3[k]$$
$$\bar{\mathbf{A}}[k+1] = \bar{\mathbf{A}}[k] + \mathbf{w}^T\left(\mathbf{M}[k] + \left(\bar{\mathbf{A}} - \hat{\mathbf{A}}[k]\right)\bar{\mathbf{F}}_{dp}[k] + \boldsymbol{\varepsilon}_p[k]\right)\bar{\mathbf{A}}'[k]$$

3 Process outputs

subscript m: measured; one sample per step

$$\delta\bar{\mathbf{p}}_{md}[k+1] = a_{\mathrm{pd}}\,\delta\bar{\mathbf{p}}_m[k] + (1-a_{\mathrm{pd}})\,\delta\bar{\mathbf{p}}_{md}[k]$$

$$\delta\bar{\mathbf{p}}_{mr}[k] = \delta\bar{\mathbf{p}}_m[k] - \delta\bar{\mathbf{p}}_{md}[k]$$

$$\mathrm{E}\big[\delta\bar{\mathbf{p}}_{mr}\,\delta\bar{\mathbf{p}}_{mr}^T\big] = C_T\,\kappa_T\,\bar{\mathbf{A}} + C_n\,\boldsymbol{\Sigma}_n$$

$$\bar{\mathbf{A}}_m[k] = \frac{\delta\bar{\mathbf{p}}_{mr}[k]\,\delta\bar{\mathbf{p}}_{mr}^T[k] - C_n\,\boldsymbol{\Sigma}_n}{C_T\,\kappa_T}$$

$$\mathrm{E}\big[\bar{\mathbf{A}}_m\big] = \bar{\mathbf{A}}$$

$$\bar{\mathbf{a}}_m[k] = [\,\bar{a}_{m,xx}\;\;\bar{a}_{m,yy}\;\;\bar{a}_{m,zz}\;\;\bar{a}_{m,xy}\;\;\bar{a}_{m,xz}\;\;\bar{a}_{m,yz}\,]^T$$

4 Angle of a gain matrix

$\theta_x = 0$

$$\bar{a}_{xx} = \bar{a}_{\parallel}\cos^2\theta_y + \bar{a}_{\perp}\sin^2\theta_y$$

$$\bar{a}_{zz} = \bar{a}_{\parallel}\sin^2\theta_y + \bar{a}_{\perp}\cos^2\theta_y$$

$$\bar{a}_{xz} = -\big(\bar{a}_{\parallel} - \bar{a}_{\perp}\big)\,\sin\theta_y\cos\theta_y$$

$$\bar{a}_{xx} - \bar{a}_{zz} = \big(\bar{a}_{\parallel} - \bar{a}_{\perp}\big)\cos 2\theta_y$$

$$-2\,\bar{a}_{xz} = \big(\bar{a}_{\parallel} - \bar{a}_{\perp}\big)\sin 2\theta_y$$

$$\tan 2\theta_y = \frac{-2\,\bar{a}_{xz}}{\bar{a}_{xx} - \bar{a}_{zz}}$$

three axes: φ_{yz} = angle of $\mathbf{w}$ in the y–z plane

$$\tan 2\theta_y = \frac{-2\,\bar{a}_{xz}}{\bar{a}_{xx} - \bar{a}_{zz}}$$

$$\tan 2\varphi_{yz} = \frac{-2\,\bar{a}_{yz}}{\bar{a}_{yy} - \bar{a}_{zz}}$$

$$\theta_x = -\arctan\big(\tan\varphi_{yz}\,\cos\theta_y\big)$$

$$\boldsymbol{\theta}(\bar{\mathbf{A}}) = [\,\theta_x\;\;\theta_y\,]^T$$

5 Computed angle

true state

$$\delta\bar{\mathbf{p}}_1[k+1] = \delta\bar{\mathbf{p}}_2[k]$$
$$\delta\bar{\mathbf{p}}_2[k+1] = \delta\bar{\mathbf{p}}_3[k]$$
$$\delta\bar{\mathbf{p}}_3[k+1] = \lambda_c\,\delta\bar{\mathbf{p}}_3[k] - \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) - \boldsymbol{\varepsilon}_p[k]$$
$$\bar{\mathbf{a}}[k+1] = \bar{\mathbf{a}}[k] + \bar{\mathbf{a}}'[k]\,\mathbf{w}^T\Big(\mathbf{M}[k] + \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) + \boldsymbol{\varepsilon}_p[k]\Big)$$
$$\bar{\mathbf{a}}'[k+1] = \bar{\mathbf{a}}'[k]$$

$$\left[\,\delta\bar{\mathbf{p}}_1^T \;\; \delta\bar{\mathbf{p}}_2^T \;\; \delta\bar{\mathbf{p}}_3^T \;\; \bar{\mathbf{a}}^T \;\; \bar{\mathbf{a}}'^T\,\right]^T \qquad (21\times 1,\ d=2)$$

process outputs, their estimates and the output matrix

$$\mathbf{y}_m[k] = \begin{bmatrix}\delta\bar{\mathbf{p}}_m[k]\\ \bar{\mathbf{a}}_m[k]\end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix}\delta\hat{\mathbf{p}}_1[k]\\ \hat{\mathbf{a}}[k]\end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix}\mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_6 & \mathbf{0}\end{bmatrix}$$

feedback: measured − estimate

$$\mathbf{e}_p[k] = \delta\bar{\mathbf{p}}_m[k] - \delta\hat{\mathbf{p}}_1[k]$$
$$\mathbf{e}_a[k] = \bar{\mathbf{a}}_m[k] - \hat{\mathbf{a}}[k]$$

estimator; $\hat{\mathbf{w}}[k]$ from $\hat{\boldsymbol{\theta}}[k] = \boldsymbol{\theta}(\hat{\mathbf{A}}[k])$

$$\hat{\mathbf{M}}[k] = \Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,\delta\hat{\mathbf{p}}_3[k]$$
$$\delta\hat{\mathbf{p}}_1[k+1] = \delta\hat{\mathbf{p}}_2[k] + \mathbf{L}_1[k]\left[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\right]^T$$
$$\delta\hat{\mathbf{p}}_2[k+1] = \delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_2[k]\left[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\right]^T$$
$$\delta\hat{\mathbf{p}}_3[k+1] = \lambda_c\,\delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_3[k]\left[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\right]^T$$
$$\hat{\mathbf{a}}[k+1] = \hat{\mathbf{a}}[k] + \hat{\mathbf{a}}'[k]\,\hat{\mathbf{w}}^T[k]\,\hat{\mathbf{M}}[k] + \mathbf{L}_4[k]\left[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\right]^T$$
$$\hat{\mathbf{a}}'[k+1] = \hat{\mathbf{a}}'[k] + \mathbf{L}_5[k]\left[\,\mathbf{e}_p^T[k] \;\; \mathbf{e}_a^T[k]\,\right]^T$$

estimator feedback matrix and error covariance

$$\mathbf{L}[k] = \mathbf{P}^f[k]\,\mathbf{H}^T\big(\mathbf{H}\,\mathbf{P}^f[k]\,\mathbf{H}^T + \mathbf{R}[k]\big)^{-1}$$
$$\mathbf{P}[k] = \big(\mathbf{I} - \mathbf{L}[k]\,\mathbf{H}\big)\,\mathbf{P}^f[k]$$

estimation error: true − estimate

$$\mathbf{e}[k] = \left[\,\big(\delta\bar{\mathbf{p}}_1 - \delta\hat{\mathbf{p}}_1\big)^T \;\; \big(\delta\bar{\mathbf{p}}_2 - \delta\hat{\mathbf{p}}_2\big)^T \;\; \big(\delta\bar{\mathbf{p}}_3 - \delta\hat{\mathbf{p}}_3\big)^T \;\; \big(\bar{\mathbf{a}} - \hat{\mathbf{a}}\big)^T \;\; \big(\bar{\mathbf{a}}' - \hat{\mathbf{a}}'\big)^T\,\right]^T$$

dynamics of estimation errors, first order, without the feedback terms

$$\delta\bar{\mathbf{p}}_1[k+1] - \delta\hat{\mathbf{p}}_1[k+1] = \delta\bar{\mathbf{p}}_2[k] - \delta\hat{\mathbf{p}}_2[k]$$

$$\delta\bar{\mathbf{p}}_2[k+1] - \delta\hat{\mathbf{p}}_2[k+1] = \delta\bar{\mathbf{p}}_3[k] - \delta\hat{\mathbf{p}}_3[k]$$

$$\delta\bar{\mathbf{p}}_3[k+1] - \delta\hat{\mathbf{p}}_3[k+1] = \lambda_c \left(\delta\bar{\mathbf{p}}_3[k] - \delta\hat{\mathbf{p}}_3[k]\right) - \mathbf{G}_a[k] \left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) - \boldsymbol{\varepsilon}_p[k]$$

$$\begin{aligned}\bar{\mathbf{a}}[k+1] - \hat{\mathbf{a}}[k+1] = {} & \left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) + \left(\bar{\mathbf{a}}'[k] - \hat{\mathbf{a}}'[k]\right) \hat{\mathbf{w}}^T[k]\, \hat{\mathbf{M}}[k] \\ & + \hat{\mathbf{a}}'[k]\, \hat{\mathbf{w}}^T[k] \Big((1-\lambda_c)\left(\delta\bar{\mathbf{p}}_3[k] - \delta\hat{\mathbf{p}}_3[k]\right) + \mathbf{G}_a[k] \left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) + \boldsymbol{\varepsilon}_p[k] \Big)\end{aligned}$$

$$\bar{\mathbf{a}}'[k+1] - \hat{\mathbf{a}}'[k+1] = \bar{\mathbf{a}}'[k] - \hat{\mathbf{a}}'[k]$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\, \mathbf{e}[k] + \begin{bmatrix} \mathbf{0} & \mathbf{0} & -\boldsymbol{\varepsilon}_p^T[k] & \left(\hat{\mathbf{a}}'[k]\, \hat{\mathbf{w}}^T[k]\, \boldsymbol{\varepsilon}_p[k]\right)^T & \mathbf{0} \end{bmatrix}^T$$

$$\mathbf{F}_e[k] = \begin{bmatrix} \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \lambda_c\, \mathbf{I}_3 & -\mathbf{G}_a[k] & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & (1-\lambda_c)\, \hat{\mathbf{a}}'[k]\, \hat{\mathbf{w}}^T[k] & \mathbf{I}_6 + \hat{\mathbf{a}}'[k]\, \hat{\mathbf{w}}^T[k]\, \mathbf{G}_a[k] & \left(\hat{\mathbf{w}}^T[k]\, \hat{\mathbf{M}}[k]\right) \mathbf{I}_6 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_6 \end{bmatrix}$$

process noise and error covariance

$$\mathbf{Q}[k] = \mathrm{E}\Big[\begin{bmatrix} \mathbf{0} & \mathbf{0} & -\boldsymbol{\varepsilon}_p^T & \left(\hat{\mathbf{a}}'\, \hat{\mathbf{w}}^T \boldsymbol{\varepsilon}_p\right)^T & \mathbf{0} \end{bmatrix}^T \begin{bmatrix} \mathbf{0} & \mathbf{0} & -\boldsymbol{\varepsilon}_p^T & \left(\hat{\mathbf{a}}'\, \hat{\mathbf{w}}^T \boldsymbol{\varepsilon}_p\right)^T & \mathbf{0} \end{bmatrix} \Big] + \mathrm{diag}\big(\mathbf{0},\ \mathbf{0},\ \mathbf{0},\ \mathbf{0},\ \mathbf{Q}_{a'}[k]\big)$$

$$\mathbf{P}^f[k+1] = \mathbf{F}_e[k]\, \mathbf{P}[k]\, \mathbf{F}_e^T[k] + \mathbf{Q}[k]$$

angle: computed from the estimate

$$\hat{\boldsymbol{\theta}}[k] = \boldsymbol{\theta}\big(\hat{\mathbf{A}}[k]\big), \qquad \hat{\mathbf{V}}[k] = \mathbf{V}\big(\hat{\boldsymbol{\theta}}[k]\big) = \begin{bmatrix} \hat{\mathbf{u}} & \hat{\mathbf{v}} & \hat{\mathbf{w}} \end{bmatrix}$$

6 Estimated angle

gain matrix from two gains and the angle; plane wall: constant angle

$$\bar{\mathbf{A}} = \mathbf{V}(\theta)\,\bar{\mathbf{\Lambda}}\,\mathbf{V}^T(\theta) = \bar{a}_{\parallel}\,\mathbf{I} + \left(\bar{a}_{\perp} - \bar{a}_{\parallel}\right)\mathbf{w}\,\mathbf{w}^T$$

$$\bar{\mathbf{A}}' = \bar{a}'_{\parallel}\left(\mathbf{I} - \mathbf{w}\,\mathbf{w}^T\right) + \bar{a}'_{\perp}\,\mathbf{w}\,\mathbf{w}^T$$

$$\hat{\mathbf{A}}[k] = \mathbf{V}\big(\hat{\boldsymbol{\theta}}[k]\big)\,\hat{\mathbf{\Lambda}}[k]\,\mathbf{V}^T\big(\hat{\boldsymbol{\theta}}[k]\big), \qquad \hat{\mathbf{\Lambda}}[k] = \mathrm{diag}\big(\hat{a}_{\parallel}[k],\ \hat{a}_{\parallel}[k],\ \hat{a}_{\perp}[k]\big)$$

first-order change of $\bar{\mathbf{A}}$; $\partial\bar{\mathbf{a}}/\partial(\cdot)$: the six elements of $\partial\bar{\mathbf{A}}/\partial(\cdot)$

$$\frac{\partial\bar{\mathbf{A}}}{\partial\bar{a}_{\parallel}} = \mathbf{I} - \mathbf{w}\,\mathbf{w}^T$$

$$\frac{\partial\bar{\mathbf{A}}}{\partial\bar{a}_{\perp}} = \mathbf{w}\,\mathbf{w}^T$$

$$\frac{\partial\bar{\mathbf{A}}}{\partial\theta_x} = \left(\bar{a}_{\perp} - \bar{a}_{\parallel}\right)\left(\frac{\partial\mathbf{w}}{\partial\theta_x}\,\mathbf{w}^T + \mathbf{w}\,\frac{\partial\mathbf{w}^T}{\partial\theta_x}\right), \qquad \frac{\partial\bar{\mathbf{A}}}{\partial\theta_y} = \left(\bar{a}_{\perp} - \bar{a}_{\parallel}\right)\left(\frac{\partial\mathbf{w}}{\partial\theta_y}\,\mathbf{w}^T + \mathbf{w}\,\frac{\partial\mathbf{w}^T}{\partial\theta_y}\right)$$

$$\frac{\partial\mathbf{w}}{\partial\boldsymbol{\theta}} = \left[\begin{array}{cc} \dfrac{\partial\mathbf{w}}{\partial\theta_x} & \dfrac{\partial\mathbf{w}}{\partial\theta_y} \end{array}\right] = \begin{bmatrix} -\sin\theta_y \sin\theta_x & \cos\theta_y \cos\theta_x \\ -\cos\theta_x & 0 \\ -\cos\theta_y \sin\theta_x & -\sin\theta_y \cos\theta_x \end{bmatrix}$$

$$\bar{\mathbf{a}} - \hat{\mathbf{a}} = \frac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_{\parallel}}\left(\bar{a}_{\parallel} - \hat{a}_{\parallel}\right) + \frac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_{\perp}}\left(\bar{a}_{\perp} - \hat{a}_{\perp}\right) + \frac{\partial\bar{\mathbf{a}}}{\partial\boldsymbol{\theta}}\left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\right)$$

true state

$$\delta\bar{\mathbf{p}}_1[k+1] = \delta\bar{\mathbf{p}}_2[k]$$

$$\delta\bar{\mathbf{p}}_2[k+1] = \delta\bar{\mathbf{p}}_3[k]$$

$$\delta\bar{\mathbf{p}}_3[k+1] = \lambda_c\,\delta\bar{\mathbf{p}}_3[k] - \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) - \boldsymbol{\varepsilon}_p[k]$$

$$\bar{a}_{\parallel}[k+1] = \bar{a}_{\parallel}[k] + \bar{a}'_{\parallel}[k]\,\mathbf{w}^T\Big(\mathbf{M}[k] + \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) + \boldsymbol{\varepsilon}_p[k]\Big)$$

$$\bar{a}_{\perp}[k+1] = \bar{a}_{\perp}[k] + \bar{a}'_{\perp}[k]\,\mathbf{w}^T\Big(\mathbf{M}[k] + \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) + \boldsymbol{\varepsilon}_p[k]\Big)$$

$$\bar{a}'_{\parallel}[k+1] = \bar{a}'_{\parallel}[k]$$

$$\bar{a}'_{\perp}[k+1] = \bar{a}'_{\perp}[k]$$

$$\boldsymbol{\theta}[k+1] = \boldsymbol{\theta}[k]$$

$$\left[\begin{array}{ccccccc} \delta\bar{\mathbf{p}}_1^T & \delta\bar{\mathbf{p}}_2^T & \delta\bar{\mathbf{p}}_3^T & \bar{a}_{\parallel} & \bar{a}_{\perp} & \bar{a}'_{\parallel} & \bar{a}'_{\perp} \end{array}\ \boldsymbol{\theta}^T\right]^T \qquad (15 \times 1,\ d = 2)$$

process outputs, their estimates and the output matrix (derivatives at the estimate)

$$\mathbf{y}_m[k] = \begin{bmatrix} \delta\bar{\mathbf{p}}_m[k] \\ \bar{\mathbf{a}}_m[k] \end{bmatrix}, \qquad \hat{\mathbf{y}}[k] = \begin{bmatrix} \delta\hat{\mathbf{p}}_1[k] \\ \hat{\mathbf{a}}[k] \end{bmatrix}$$

$$\mathbf{H}[k] = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_{\parallel}} & \dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_{\perp}} & \mathbf{0} & \mathbf{0} & \dfrac{\partial\bar{\mathbf{a}}}{\partial\boldsymbol{\theta}} \end{bmatrix}$$

feedback: measured − estimate

$$\mathbf{e}_p[k] = \delta\bar{\mathbf{p}}_m[k] - \delta\hat{\mathbf{p}}_1[k]$$

$$\mathbf{e}_a[k] = \bar{\mathbf{a}}_m[k] - \hat{\mathbf{a}}[k]$$

estimator; $\hat{\mathbf{w}}[k]$ from the estimate $\hat{\boldsymbol{\theta}}[k]$; $\mathbf{L}[k]$, $\mathbf{P}[k]$ as in Section 5

$$
\begin{aligned}
\hat{\mathbf{M}}[k] &= \Delta\bar{\mathbf{p}}_d[k] + (1-\lambda_c)\,\delta\hat{\mathbf{p}}_3[k] \\
\delta\hat{\mathbf{p}}_1[k+1] &= \delta\hat{\mathbf{p}}_2[k] + \mathbf{L}_1[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T \\
\delta\hat{\mathbf{p}}_2[k+1] &= \delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_2[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T \\
\delta\hat{\mathbf{p}}_3[k+1] &= \lambda_c\,\delta\hat{\mathbf{p}}_3[k] + \mathbf{L}_3[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T \\
\hat{a}_\parallel[k+1] &= \hat{a}_\parallel[k] + \hat{a}'_\parallel[k]\,\hat{\mathbf{w}}^T[k]\,\hat{\mathbf{M}}[k] + \mathbf{L}_4[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T \\
\hat{a}_\perp[k+1] &= \hat{a}_\perp[k] + \hat{a}'_\perp[k]\,\hat{\mathbf{w}}^T[k]\,\hat{\mathbf{M}}[k] + \mathbf{L}_5[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T \\
\hat{a}'_\parallel[k+1] &= \hat{a}'_\parallel[k] + \mathbf{L}_6[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T \\
\hat{a}'_\perp[k+1] &= \hat{a}'_\perp[k] + \mathbf{L}_7[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T \\
\hat{\boldsymbol{\theta}}[k+1] &= \hat{\boldsymbol{\theta}}[k] + \mathbf{L}_8[k]\begin{bmatrix}\mathbf{e}_p^T[k] & \mathbf{e}_a^T[k]\end{bmatrix}^T
\end{aligned}
$$

estimation error: true − estimate

$$
\mathbf{e}[k] = \begin{bmatrix}\left(\delta\bar{\mathbf{p}}_1 - \delta\hat{\mathbf{p}}_1\right)^T & \left(\delta\bar{\mathbf{p}}_2 - \delta\hat{\mathbf{p}}_2\right)^T & \left(\delta\bar{\mathbf{p}}_3 - \delta\hat{\mathbf{p}}_3\right)^T & \bar{a}_\parallel - \hat{a}_\parallel & \bar{a}_\perp - \hat{a}_\perp & \bar{a}'_\parallel - \hat{a}'_\parallel & \bar{a}'_\perp - \hat{a}'_\perp & \left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\right)^T\end{bmatrix}^T
$$

dynamics of estimation errors, first order, without the feedback terms

$$
\begin{aligned}
\delta\bar{\mathbf{p}}_1[k+1] - \delta\hat{\mathbf{p}}_1[k+1] &= \delta\bar{\mathbf{p}}_2[k] - \delta\hat{\mathbf{p}}_2[k] \\
\delta\bar{\mathbf{p}}_2[k+1] - \delta\hat{\mathbf{p}}_2[k+1] &= \delta\bar{\mathbf{p}}_3[k] - \delta\hat{\mathbf{p}}_3[k] \\
\delta\bar{\mathbf{p}}_3[k+1] - \delta\hat{\mathbf{p}}_3[k+1] &= \lambda_c\left(\delta\bar{\mathbf{p}}_3[k] - \delta\hat{\mathbf{p}}_3[k]\right) - \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) - \boldsymbol{\varepsilon}_p[k] \\
\bar{a}_\parallel[k+1] - \hat{a}_\parallel[k+1] &= \left(\bar{a}_\parallel[k] - \hat{a}_\parallel[k]\right) + \left(\bar{a}'_\parallel[k] - \hat{a}'_\parallel[k]\right)\hat{\mathbf{w}}^T[k]\,\hat{\mathbf{M}}[k] + \hat{a}'_\parallel[k]\,\hat{\mathbf{M}}^T[k]\,\frac{\partial\mathbf{w}}{\partial\boldsymbol{\theta}}\left(\boldsymbol{\theta}[k] - \hat{\boldsymbol{\theta}}[k]\right) \\
&\quad + \hat{a}'_\parallel[k]\,\hat{\mathbf{w}}^T[k]\Big((1-\lambda_c)\left(\delta\bar{\mathbf{p}}_3[k] - \delta\hat{\mathbf{p}}_3[k]\right) + \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) + \boldsymbol{\varepsilon}_p[k]\Big) \\
\bar{a}_\perp[k+1] - \hat{a}_\perp[k+1] &= \left(\bar{a}_\perp[k] - \hat{a}_\perp[k]\right) + \left(\bar{a}'_\perp[k] - \hat{a}'_\perp[k]\right)\hat{\mathbf{w}}^T[k]\,\hat{\mathbf{M}}[k] + \hat{a}'_\perp[k]\,\hat{\mathbf{M}}^T[k]\,\frac{\partial\mathbf{w}}{\partial\boldsymbol{\theta}}\left(\boldsymbol{\theta}[k] - \hat{\boldsymbol{\theta}}[k]\right) \\
&\quad + \hat{a}'_\perp[k]\,\hat{\mathbf{w}}^T[k]\Big((1-\lambda_c)\left(\delta\bar{\mathbf{p}}_3[k] - \delta\hat{\mathbf{p}}_3[k]\right) + \mathbf{G}_a[k]\left(\bar{\mathbf{a}}[k] - \hat{\mathbf{a}}[k]\right) + \boldsymbol{\varepsilon}_p[k]\Big) \\
\bar{a}'_\parallel[k+1] - \hat{a}'_\parallel[k+1] &= \bar{a}'_\parallel[k] - \hat{a}'_\parallel[k] \\
\bar{a}'_\perp[k+1] - \hat{a}'_\perp[k+1] &= \bar{a}'_\perp[k] - \hat{a}'_\perp[k] \\
\boldsymbol{\theta}[k+1] - \hat{\boldsymbol{\theta}}[k+1] &= \boldsymbol{\theta}[k] - \hat{\boldsymbol{\theta}}[k]
\end{aligned}
$$

$$
\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] + \begin{bmatrix}\mathbf{0} & \mathbf{0} & -\boldsymbol{\varepsilon}_p^T[k] & \hat{a}'_\parallel[k]\,\hat{\mathbf{w}}^T[k]\,\boldsymbol{\varepsilon}_p[k] & \hat{a}'_\perp[k]\,\hat{\mathbf{w}}^T[k]\,\boldsymbol{\varepsilon}_p[k] & 0 & 0 & \mathbf{0}\end{bmatrix}^T
$$

$$
\mathbf{F}_e[k] = \begin{bmatrix}
\mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & \lambda_c\,\mathbf{I}_3 & -\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_\parallel} & -\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_\perp} & \mathbf{0} & \mathbf{0} & -\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\boldsymbol{\theta}} \\
\mathbf{0} & \mathbf{0} & (1-\lambda_c)\,\hat{a}'_\parallel\,\hat{\mathbf{w}}^T & 1+\hat{a}'_\parallel\,\hat{\mathbf{w}}^T\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_\parallel} & \hat{a}'_\parallel\,\hat{\mathbf{w}}^T\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_\perp} & \hat{\mathbf{w}}^T\hat{\mathbf{M}} & 0 & \hat{a}'_\parallel\,\hat{\mathbf{w}}^T\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\boldsymbol{\theta}} + \hat{a}'_\parallel\,\hat{\mathbf{M}}^T\dfrac{\partial\mathbf{w}}{\partial\boldsymbol{\theta}} \\
\mathbf{0} & \mathbf{0} & (1-\lambda_c)\,\hat{a}'_\perp\,\hat{\mathbf{w}}^T & \hat{a}'_\perp\,\hat{\mathbf{w}}^T\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_\parallel} & 1+\hat{a}'_\perp\,\hat{\mathbf{w}}^T\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\bar{a}_\perp} & 0 & \hat{\mathbf{w}}^T\hat{\mathbf{M}} & \hat{a}'_\perp\,\hat{\mathbf{w}}^T\mathbf{G}_a\dfrac{\partial\bar{\mathbf{a}}}{\partial\boldsymbol{\theta}} + \hat{a}'_\perp\,\hat{\mathbf{M}}^T\dfrac{\partial\mathbf{w}}{\partial\boldsymbol{\theta}} \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 1 & 0 & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & 0 & 0 & 0 & 1 & \mathbf{0} \\
\mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_2
\end{bmatrix}
$$

process noise and error covariance

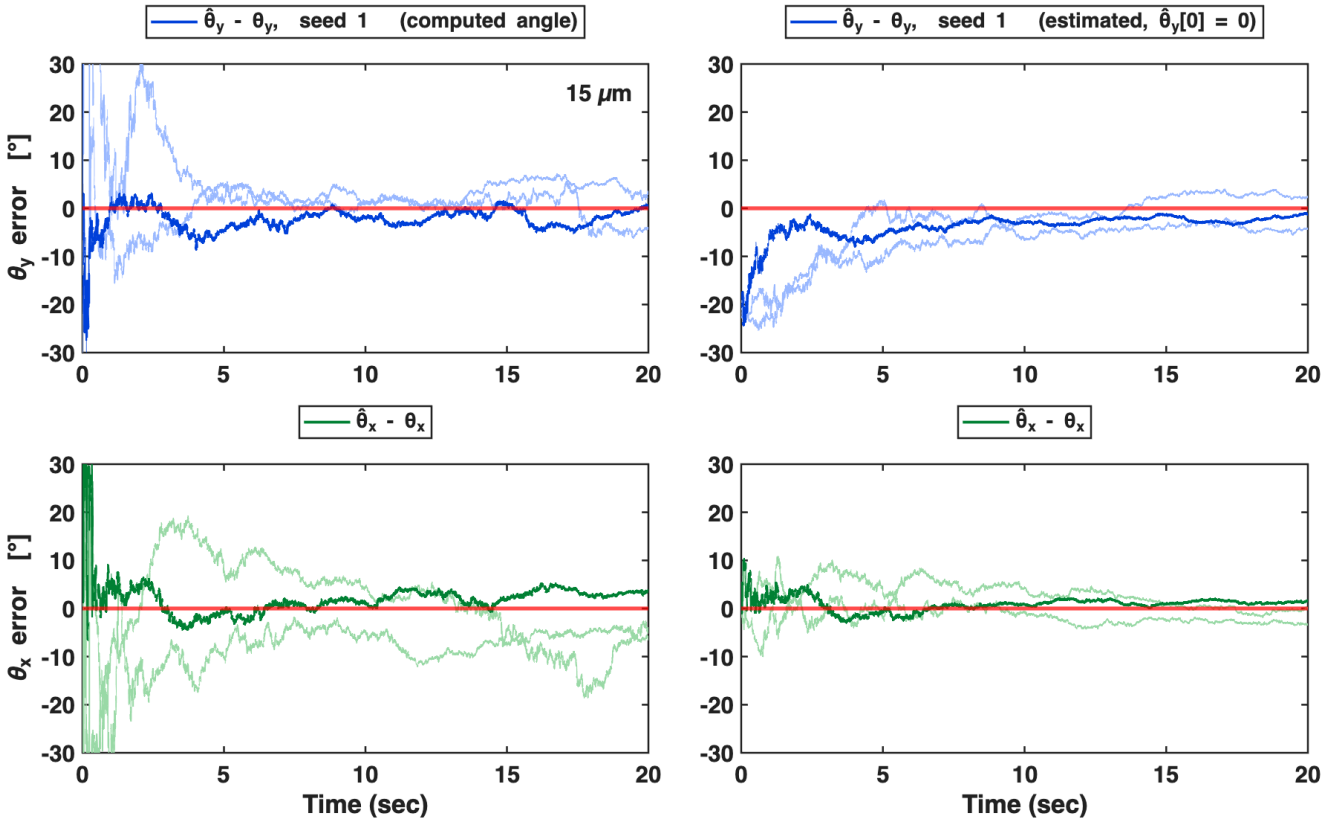
$$
\begin{aligned}
\mathbf{Q}[k] &= \mathrm{E}\Big[\begin{bmatrix}\mathbf{0} & \mathbf{0} & -\boldsymbol{\varepsilon}_p^T & \hat{a}'_\parallel\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p & \hat{a}'_\perp\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p & 0 & 0 & \mathbf{0}\end{bmatrix}^T\begin{bmatrix}\mathbf{0} & \mathbf{0} & -\boldsymbol{\varepsilon}_p^T & \hat{a}'_\parallel\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p & \hat{a}'_\perp\,\hat{\mathbf{w}}^T\boldsymbol{\varepsilon}_p & 0 & 0 & \mathbf{0}\end{bmatrix}\Big] \\
&\quad + \mathrm{diag}\big(\mathbf{0},\ \mathbf{0},\ \mathbf{0},\ 0,\ 0,\ \mathbf{Q}_{a'}[k],\ \mathbf{0}\big) \\
\mathbf{P}^f[k+1] &= \mathbf{F}_e[k]\,\mathbf{P}[k]\,\mathbf{F}_e^T[k] + \mathbf{Q}[k]
\end{aligned}
$$

angle: the estimate

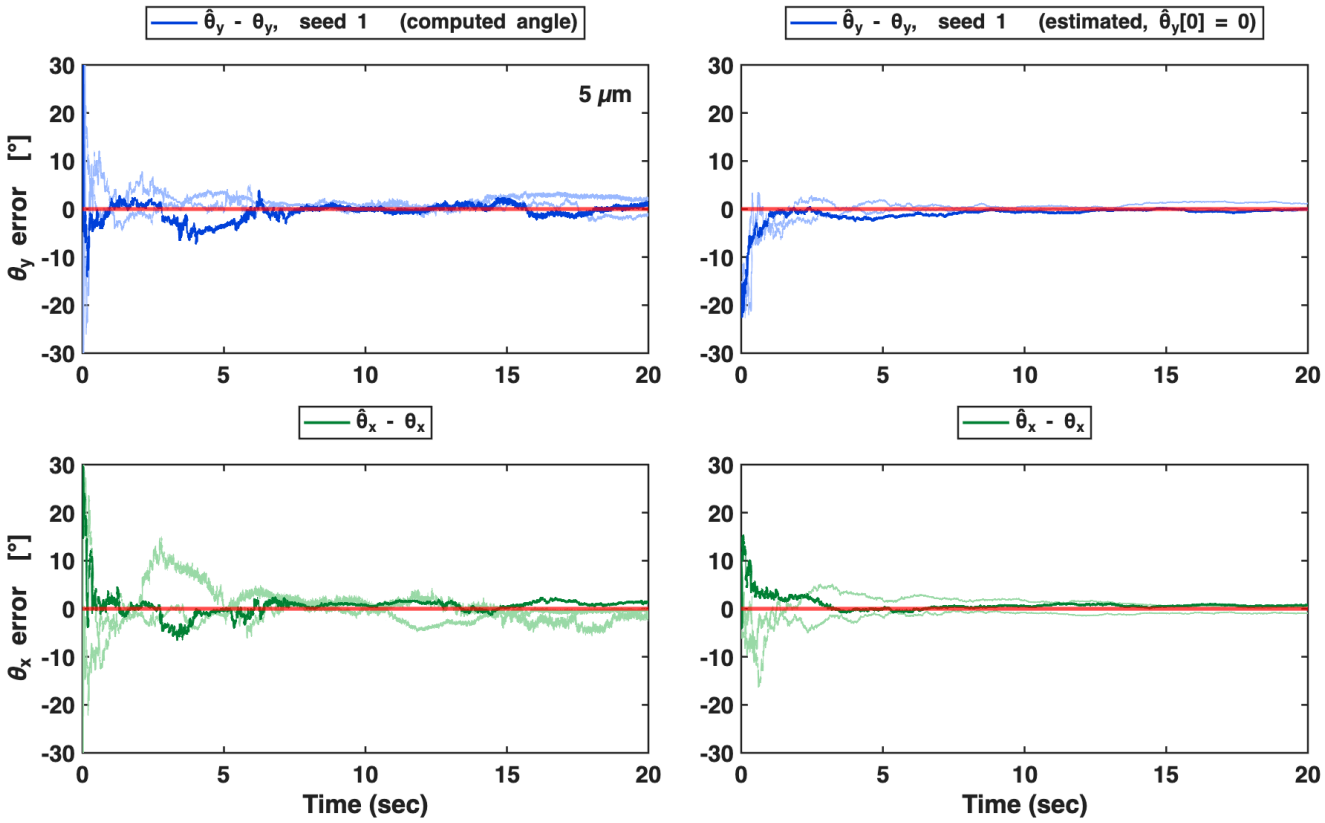
$$
\hat{\mathbf{V}}[k] = \mathbf{V}\big(\hat{\boldsymbol{\theta}}[k]\big) = [\,\hat{\mathbf{u}}\ \ \hat{\mathbf{v}}\ \ \hat{\mathbf{w}}\,]
$$

7 Fixed-point results

wall $\theta_y = 20°$, $\theta_x = 0$; probe held at 15 µm (top) and 5 µm (bottom); left computed angle, right estimated angle with $\hat{\boldsymbol{\theta}}[0] = \mathbf{0}$; seed 1 dark, seeds 2–3 light



Angle errors at 15 µm over 20 s.

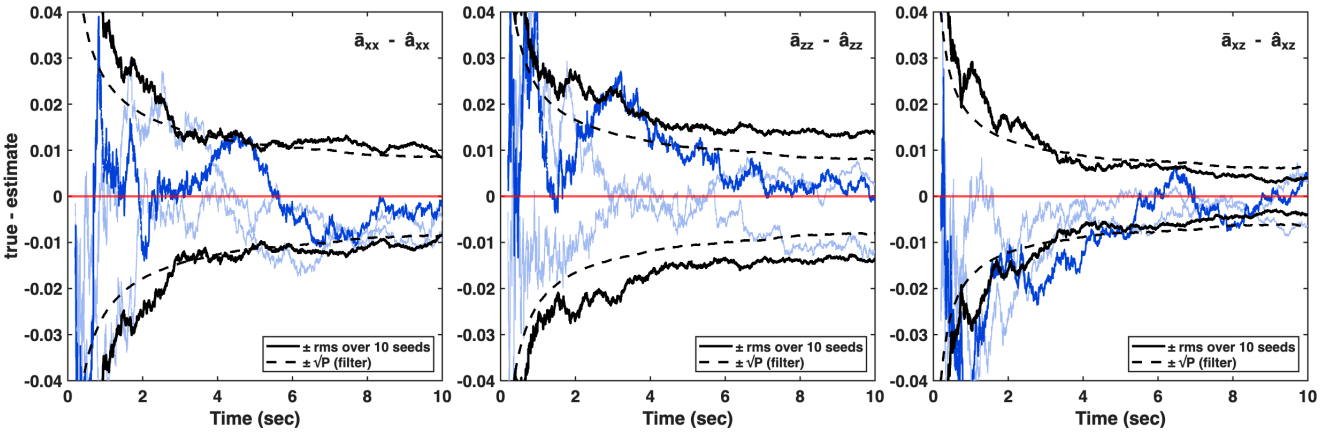


Angle errors at 5 µm over 20 s.

8 Angle error of the computed angle

The computed angle is the angle formula evaluated at $\hat{\mathbf{A}}$, so its error is the error of $\hat{\mathbf{A}}$ times the sensitivity of the formula. (1) $\hat{\mathbf{A}}$: the errors of the three elements in the formula are of the same size at 15 and 5 µm. (2) Sensitivity: $1/[2(\bar{a}_{\parallel} - \bar{a}_{\perp})]$, larger farther from the wall, so the same error of $\hat{\mathbf{A}}$ gives a larger angle error and a slower convergence at 15 µm.

15 µm; lines: 3 seeds with the smallest error; black: ±rms of 10 seeds (solid), $\pm\sqrt{P}$ (dashed)



angle from the gain matrix (Section 4)

$$\tan 2\theta_y = \frac{N}{D}$$

$$N = -2\,\bar{a}_{xz} = (\bar{a}_{\parallel} - \bar{a}_{\perp}) \sin 2\theta_y$$

$$D = \bar{a}_{xx} - \bar{a}_{zz} = (\bar{a}_{\parallel} - \bar{a}_{\perp}) \cos 2\theta_y$$

sensitivity of the angle to N and D

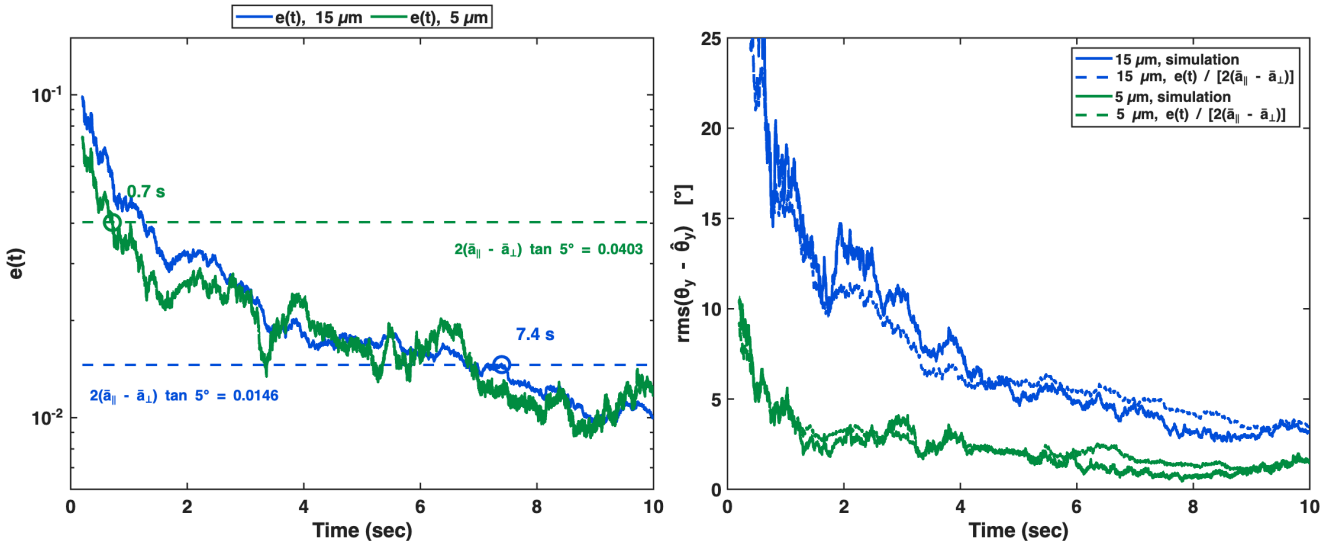
$$\frac{\partial \theta_y}{\partial N} = \frac{\cos 2\theta_y}{2(\bar{a}_{\parallel} - \bar{a}_{\perp})}, \qquad \frac{\partial \theta_y}{\partial D} = \frac{-\sin 2\theta_y}{2(\bar{a}_{\parallel} - \bar{a}_{\perp})}$$

$e_N = N - \hat{N}$, $e_D = D - \hat{D}$; $e(t)$: rms of e_N, e_D

$$\theta_y - \hat{\theta}_y \approx \frac{\cos 2\theta_y\, e_N - \sin 2\theta_y\, e_D}{2(\bar{a}_{\parallel} - \bar{a}_{\perp})}$$

$$\mathrm{rms}(\theta_y - \hat{\theta}_y) \approx \frac{e(t)}{2(\bar{a}_{\parallel} - \bar{a}_{\perp})}$$

$\bar{a}_\parallel - \bar{a}_\perp = 0.083$ (15 µm), 0.230 (5 µm); left: $e(t)$, dashed $2(\bar{a}_\parallel - \bar{a}_\perp)\tan 5°$; right: simulation, dashed $e(t)/[2(\bar{a}_\parallel - \bar{a}_\perp)]$



Same error size $e(t)$ at both heights; the angle error is $e(t)/[2(\bar{a}_\parallel - \bar{a}_\perp)]$.

9 Angle error of the estimated angle

The estimated angle reads the angle from the same elements through the angle columns of $\mathbf{H}$, whose size is $2(\bar{a}_{\parallel} - \bar{a}_{\perp})$. (1) The angle information per step is proportional to $(\bar{a}_{\parallel} - \bar{a}_{\perp})^2$, so $\sqrt{P_{\theta_y}}$ follows the same $e(t)/[2(\bar{a}_{\parallel} - \bar{a}_{\perp})]$ as the computed angle. (2) In addition, the initial value is kept with the weight $P_{\theta_y}(t)/P_{\theta_y}(0)$: with little information the estimate stays near the initial value instead of jumping.

angle columns of $\mathbf{H}$ for N and D (Section 6)

$$\frac{\partial N}{\partial \theta_y} = 2\big(\bar{a}_{\parallel} - \bar{a}_{\perp}\big)\cos 2\theta_y, \qquad \frac{\partial D}{\partial \theta_y} = -2\big(\bar{a}_{\parallel} - \bar{a}_{\perp}\big)\sin 2\theta_y$$

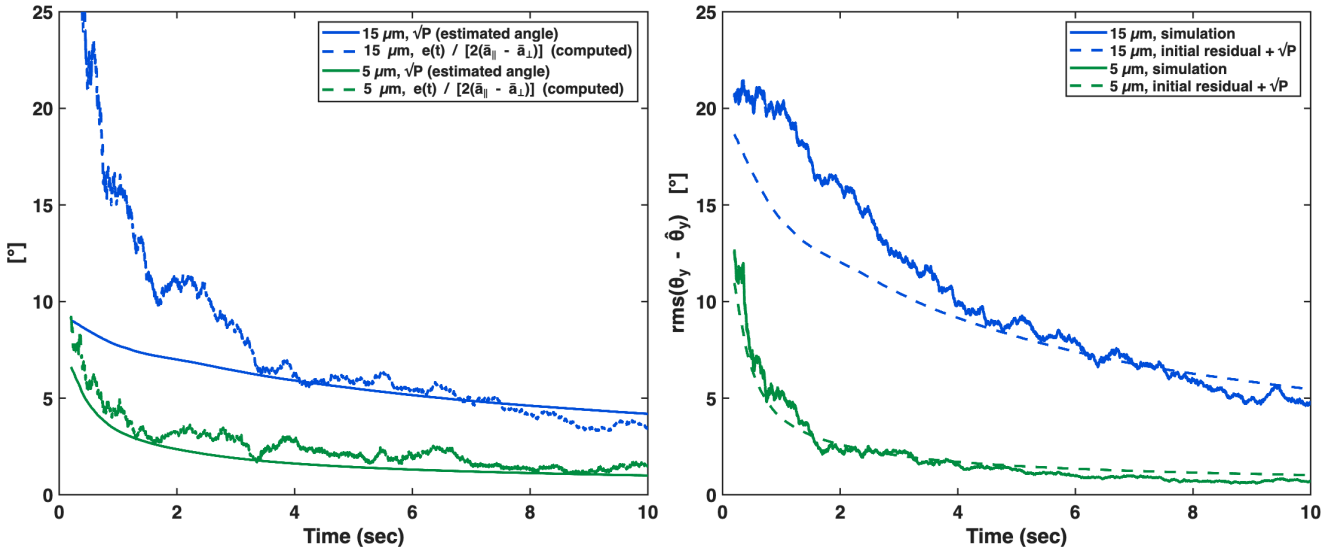
uncertainty of the estimated angle

$$\sqrt{P_{\theta_y}(t)} \approx \frac{e(t)}{2\big(\bar{a}_{\parallel} - \bar{a}_{\perp}\big)}$$

error: initial residual and noise

$$\theta_y - \hat{\theta}_y \approx \frac{P_{\theta_y}(t)}{P_{\theta_y}(0)}\big(\theta_y - \hat{\theta}_y(0)\big) + \text{noise of size } \sqrt{P_{\theta_y}(t)}$$

$\hat{\theta}_y(0) = 0$, $\sqrt{P_{\theta_y}(0)} = 10°$, wall 20°; 15 and 5 µm, seeds 1–10; left: $\sqrt{P_{\theta_y}}$ (solid), $e(t)/[2(\bar{a}_{\parallel} - \bar{a}_{\perp})]$ of the computed angle (dashed); right: simulation (solid), initial residual and $\sqrt{P_{\theta_y}}$ (dashed)



Same information limit as the computed angle; the remaining error is the initial value.